

EVAN VIENNEAU

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University of Alberta, Edmonton, Alberta, Canada

RESEARCH INTERESTS

Dark matter, computational astrophysics, indirect detection, direct detection, black holes, quantum gravity

EDUCATION

Doctor of Philosophy (Ph.D) Physics

University of Alberta

September 2021 - Present

Edmonton, AB

- Supervised by Prof. Nassim Bozorgnia and Prof. Saeed Rastgoo
- Using cosmological simulations to study the distribution and dynamics of dark matter in the context of direct and indirect detection, for a variety of astrophysical targets and dark matter models
- Using numerical methods to derive characteristics of quantum-corrected black holes modeling singularity resolution
- Anticipated completion in Spring 2026

Honours Bachelor of Physics Co-op

McMaster University

September 2016 - April 2021

Hamilton, ON

- Minor in computer science
- Senior thesis entitled “Properties of Low Mass Disk-Dominated Galaxies in Voids, Filaments, and Groups”

HONOURS & AWARDS

UofA Department of Physics Ph.D Research Award

2025

Alberta Graduate Excellence Scholarship

2025

First Place Student Talk in Particle Physics Division at 2024 CAP Conference

2024

PUBLICATIONS

1. F. Fragomeno, S. Hergott, S. Rastgoo, **E. Vienneau**, *A rotating GUP black hole: metric, shadow, and bounds on quantum parameters*, [arXiv:2603.23680 \[gr-qc\]](https://arxiv.org/abs/2603.23680) Submitted for publication in JCAP
2. **E. Vienneau**, E. Batteas, A. J. Evans, O. V. Hartl, N. Bozorgnia, L. E. Strigari, *Dark matter annihilation signals from the Large Magellanic Cloud and its impact on the Milky Way*, [arXiv:2509.13540 \[astro-ph.HE\]](https://arxiv.org/abs/2509.13540) Accepted for publication in JCAP
3. O. V. Hartl, **E. Vienneau**, E. Batteas, A. J. Evans, N. Bozorgnia, L. E. Strigari, *Enhancements in velocity-dependent dark matter annihilation in Galactic subhalos*, [arXiv:2509.05519 \[astro-ph.HE\]](https://arxiv.org/abs/2509.05519) Accepted for publication in JCAP
4. E. Pinetti, **E. Vienneau**, N. Bozorgnia, *Dark matter decay signals in cosmic filaments*, [arXiv:2504.08025 \[astro-ph.CO\]](https://arxiv.org/abs/2504.08025) Submitted to PRL
5. F. Fragomeno, D. M. Gingrich, S. Hergott, S. Rastgoo, **E. Vienneau**, *A generalized uncertainty-inspired quantum black hole*, *Phys.Rev.D* 111, 024048 (2025), [arXiv:2506.03909 \[gr-qc\]](https://arxiv.org/abs/2506.03909)
6. **E. Vienneau**, A. J. Evans, O. V. Hartl, N. Bozorgnia, L. E. Strigari, A. H. Riley, N. Shipp, *Significant impact of Galactic dark matter particles on annihilation signals from Sagittarius analogues*, *JCAP* 10 (2024) 019, [arXiv:2403.15544 \[astro-ph.HE\]](https://arxiv.org/abs/2403.15544)

7. E. Rahimi, **E. Vienneau**, N. Bozorgnia *The local dark matter distribution in self-interacting dark matter halos*, JCAP 02 (2023) 040, [arXiv:2210.06498](https://arxiv.org/abs/2210.06498) [astro-ph.CO]
8. T. Brown,..., **E. Vienneau**, et al., *VERTICO: The Virgo Environment Traced In CO Survey*, ApJS 257 21, [arXiv:2111.00937](https://arxiv.org/abs/2111.00937) [astro-ph.GA]

INVITED PRESENTATIONS

Searching for dark matter annihilation and decay: From dwarfs to cosmic filaments December 2025
High Energy Phenomenology Journal Club *Texas A&M University, USA*

PRESENTATIONS

Impact of Galactic dark matter on annihilation signals from Sagittarius November 2025
TeV Particle Astrophysics Conference *Fundación Bancaja, Spain*

How does the LMC impact dark matter annihilation signals in the MW? August 2025
New Approaches to dark matter in galaxies and the Local Group Workshop *BIRS, Alberta*

Searching for dark matter decay in cosmic filaments April 2025
Astrophysics Seminar *University of Alberta, Canada*

Impact of the LMC on DM indirect detection May 2024
2024 Canadian Association of Physicists Congress *Western University, Canada*

- First place student talk in particle physics division

A generalized uncertainty-inspired quantum black hole May 2024
Loops'24 International Conference on Quantum Gravity *Florida Atlantic University, USA*

The local dark matter distribution in self-interacting dark matter halos June 2023
2023 Canadian Association of Physicists Congress *University of New Brunswick, Canada*

Enhanced star formation from ram-pressure stripping in Virgo galaxies December 2020
VERTICO Collaboration Meeting *McMaster University, Canada*

Properties of disk-dominated galaxies in voids, filaments, and groups May 2020
Department of Physics and Astronomy Poster Fair *McMaster University, Canada*

Calibration source umbilical behaviour in linear alkylbenzene March 2019
SNO+ Bi-Yearly Collaboration Meeting *SNOLAB*

SCIENTIFIC VISITS

2023 Canadian Astroparticle Physics Community Meeting August 2023
QueensU *Kingston, Ontario*

- Attended a variety of presentations, focused on communicating the research progress achieved by the Canadian astroparticle physics community

2022 Tri-Institute Summer School on Elementary Particles July 2022
TRIUMF *Vancouver, British Columbia*

- Attended a diverse array of lectures to learn, engage and network with leading experts in the field

WORK EXPERIENCE

Astrophysics Researcher and Beta Tester September 2020 - April 2021
McMaster University *Hamilton, ON*

- Acted as beta-tester for the development of a state-of-the-art cloud-based computing environment named ARCADE
- Worked with an ALMA imaging pipeline to optimize the processing of interferometric radio continuum data from the VERTICO survey of galaxies
- Analysis of this data lead to potential evidence for enhanced radio continuum emission on the leading edge of infalling Virgo cluster galaxies

Physics Pedagogy Researcher

McMaster University

July 2020 - August 2020

Hamilton, ON

- Developed web-based Python programming bootcamp for incoming science undergraduates
- Investigated ways to better incorporate computation into the undergraduate physics curriculum
- Supported instructors in the organizational and logistical side of running online courses

Neutrino Detection Researcher

SNOLAB

January 2019 - June 2019

Lively, ON

- Ran quality assurance tests on scintillator samples using density, turbidity and UV absorption meters
- Monitored state of cover gas bags in detector cavity to maintain correct nitrogen levels
- Worked detector operator shifts monitoring and gathering detector data
- Performed independent research on the performance of calibration source umbilicals in scintillator fluid

OTHER RESPONSIBILITIES

Graduate Student Physics Association - Ambassador

University of Alberta

January 2024 - December 2024

Edmonton, AB

- Acted as the main liaison between the graduate student physics association and the graduate student association

Graduate Student Physics Association - VP Finance

University of Alberta

January 2023 - December 2023

Edmonton, AB

- Oversaw GPSA finances and managed GPSA bank account
- Supported fundraising activities

Physics Graduate Student Association - YUGSA representative

York University

September 2021 - December 2021

Toronto, ON

- Networking and community development

TEACHING ASSISTANTSHIP

PHYS130 - Wave Motion, Optics and Sound

University of Alberta

September 2025 - December 2025

Edmonton, AB

- Lead TA focused on improving labs, mentoring TAs and identifying areas of improvement in labs
- Running and marking of lab sessions

PHYS144 - Newtonian Mechanics

University of Alberta

September 2024 - December 2024

Edmonton, AB

- Running and marking of lab sessions

PHYS146 - Fluids and Waves

University of Alberta

January 2024 - April 2024

Edmonton, AB

- Overseeing and facilitating semester long student research projects in a variety of physics subfields

ENPHYS131 - Mechanics*University of Alberta*

January 2023 - April 2023

Edmonton, AB

- Running and marking of lab sessions

PHYS130 - Wave Motion, Optics and Sound*University of Alberta*

September 2022 - December 2022

Edmonton, AB

- Running of large class tutorial sessions and marking

PHYS124 - Particles and Waves*University of Alberta*

September 2022 - December 2022

Edmonton, AB

- Running of large class tutorial sessions and marking

PHYS1421 - Physics with Life Science Applications 1*York University*

May 2022 - June 2022

Toronto, ON

- Invigilation of midterms and exams
- Marking of class assignments, midterms and exam

PHYS2030 - Computational Methods for Physicists and Engineers*York University*

January 2022 - April 2022

Toronto, ON

- Marking of weekly assignments

PHYS1422 - Physics with Life Science Applications 2*York University*

January 2022 - April 2022

Toronto, ON

- Marking of class tests

PHYS1411 - Physics Fundamentals 1*York University*

September 2021 - December 2021

Toronto, ON

- Running of lab tutorial sessions and marking labs